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Subject Name: Database Management System

Week 1

Module 1 : Introduction to Database

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Lecture No: 1:Introduction to database systems, File Vs. Database system



Introduction to Database Management System

- defi :-A DBMS is a software system that helps in process. of defining, constructing, manipulating the database. with providing Authentication, Privacy & security to Database software's eg- DB2 by IBM, MySOL, Oracle, FoxPro by Microsoft.
- DBMS has become integral part of the information system (IS) many organizations as it's used to handle huge amount of data. we can say as the core Module of the IS is DBMS.
- before going in detailed about DBMS; we see the data; information database details

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- Data :- defi- The facts & Figures that can be recorded in system & that have some special meaning that assigned to it.
- The system can manual (register) system or computerized system eg- Data of a customer like name, telephone Number Address, & purchase date of product. Etc
- In simple words we can say that data is nothing but row data. When we process this row data then we can have some information": eg - In which month company has more sale [after processing /Applying Algorithms 1 years data of company]· or How many students are there in first 10 ranks From out our class. [After processing Marks. data of whole class]

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- In simple, we can say that information is nothing but Fruitful information extracted from raw data. This information is useful for understanding the results of different types of data.
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- **Students_infor**---→

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- **Course_infor**---→

Cid	Name	Hours

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File system vs DBMS system

DBMS	File Processing System
Data redundancy is minimal	Data redundancy is high.
Data inconsistency is minimized or does not occur under normal conditions	Data inconsistency may occur due to duplicate data.
Accessing database is easier	Accessing is comparatively difficult
Data isolation is minimized because all data is stored in a centralized database.	Data is scattered across multiple files in different formats, leading to data isolation
Transactions like insert, delete, view, updating, etc are possible in database	In file system, transactions are not possible
Concurrent access and recovery is possible in database	Concurrent access and recovery is not possible
Security of data	Security of data is not good
Relationships among data are maintained using tables, keys, and constraints. Managed by a Database Management System (DBMS).	Relationships are maintained manually by application programs or file structures. No automatic relationship management.

Lecture No: 2: Characteristics of DBMS, role of DBA, Sills required for DBA



Characteristics of DBMS

1. Data Integrity
2. Data Security
3. Data independence
4. Transaction Control
5. Concurrency control
6. Data recovery



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1. Data Integrity

- Integrity Constraints provides a way of ensuring that changes made to database by authorized users that do not result in a loss of data consistency & correctness.
- Database integrity concern with the correctness & completeness of data in the database.
- This objective can never be guaranteed, one cannot ensure that every entry made in database is accurate.
- Example: a user can accidently try to enter a phone number into Date field



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2. Data Security

- A DBMS has separate system for security which is responsible for protecting database against accidental or destructions or misuse. intentional loss.
- Data in database should be given to authorized users & authorised user's should have access to write/ modify the data at any time, anywhere.
- In simple words, Data security refers to the data prevention. of data from unauthorized users. it's only allowed to access the data to authorized users.
- Data security can be implemented through user accounts (passwords) & authentication schemes.

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3. Data independence

- it is defined as the capacity to change data kept at one place without changing data kept at other locations.
- or it means that the ability to modify the schema definition in one level should not affect the schema definition in the next higher level.
- schema - the basic structure of database i.e. how the data will be stored in database. & Instance - it's set of information stored at particular time.

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4. Transaction Control - (roll back)..

- Transactions are set of operations used to perform a logical set of work.

A transaction usually means that the data in database has changed.

- One of the major uses of the DBMS is to protect the data from system failures. it's done by ensuring that all the data is restored to a consistent state when computer is restarted after a crash.
- The transaction is program in a DBMS. Executing the same program multiple times will generate multiple transactions.
- The changes made to database can be with the help of roll back command. & The changes can be saved with successful with help of commit data command.



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- The changes made to database can be with the help of roll back command. & The changes can be saved with successful with help of commit data command.

5) concurrency control :

- The data in database can be accessed by multiple users at same point of time. Such operations allowed by sharing same data between multiple users.
- In simple, Data concurrency is the ability to allow multiple users to affect multiple transactions within a database

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- data concurrency allows multiple users to access data all at the same time. The ability to offer concurrency is unique
- If many transactions try to access the same data, then in consistency arises. Concurrency control required to maintain consistency data.
- for example, if we take ATM machines & do not use concuthen rrancy multiple persons cannot draw money at a time in different places. This is where we need concurrency.
- Advantages

a) Waiting time will be decreased

b) Response time will decrease

c) Resource utilization will increase

d) system performance & efficiency is increased

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6) Data recovery - backups & restore -

- Database recovery is the process of restoring the database to original (correct) state after database failure.
- The main element of database recovery is the most. recent database backup.
- If you maintain database back up efficiently, then data base is very straight forward recovery process.

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- Why ??: we need a data recovery solution if we experience inaccessibility of files & folders; unrecognized format, virus attacks, accidental deletions, permanent deletion of files inaccessible partitions, corrupt documents, unexpected system shutdowns, error messages while opening a file & likewise there could be endless reasons. where Data recovery plays very important role.
- Four Data recovery techniques are as follows:a) Mirroring

  b) Recovery using back up c) Recovery using Transaction logs

  d) shadow paging.

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Role of DBA (DataBase Administrator) :

- The DBA needs to perform many roles to keep database updated & running
- System Administrator / Designer
- The DBA needs to manage DBMS software &server.
- DBA responsible for deciding the storage & access methods
- DBA performs all data field updates or adding field into database..
- DBA also implements searching strategy.
- DBA designs database & periodically updating performed
- System Analyst
- DBA need to take care of crashes & recovery procedures.
- DBA specify the technique to monitor performance of DB
- DBA needs to analyse the system performance & Fine the DBMS activities.
- DBA needs to design DB schema.. & select H/W & S/W.

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skills required for DBA:

- There are various programming & soft skills are Required to DBA are as follows:

✓ Good communication skills

✓ Excellent knowledge of database Architecture design & RDBMS

✓ Knowledge of structured query Language [SQL]

- In addition this aspect of database administration includes maintenance of data security, which involves maintaining security authorization tables, conducting periodic security audits, investing all security breaches.



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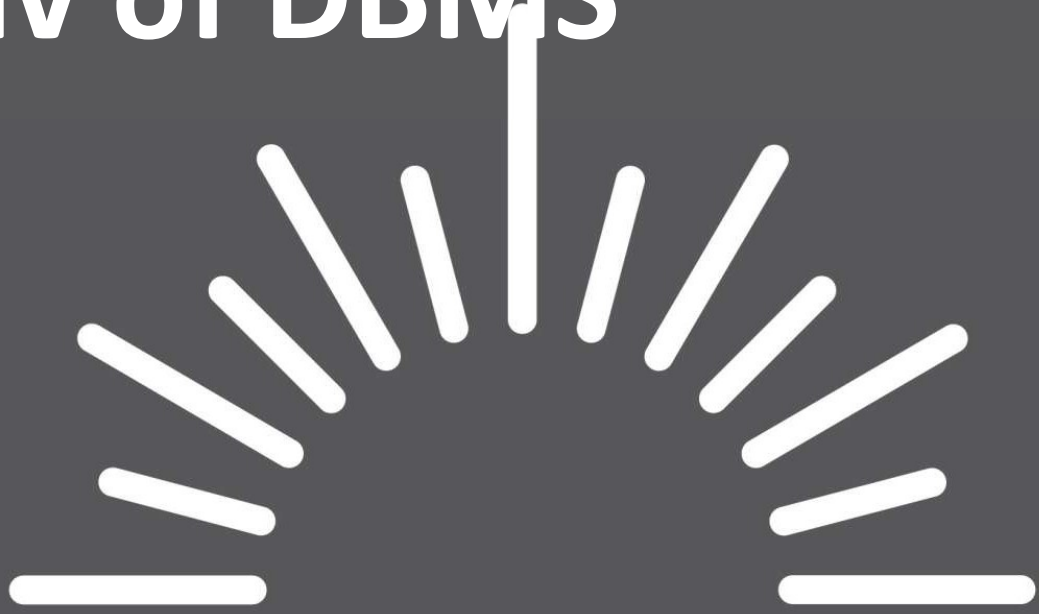
- DBA has to manage / handle all accurate information about companies data readily on hand, for this purpose be maintains "data dictionary.“
- Data dictionary contains definitions of all data items & structures, the various schemes, the relevant authorization & validation checks & different mapping definitions.
- It should also have information about source & definition of data items & flow of data item as it is used by system.
- This type of information is great help to the DBA in maintaining centralised control of data.

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- DBMS means: DBMS is a software program that is used to handle (insert, delete, update, Manipulate, create etc.) data, with providing Authenticity; Privacy; Security.. eg → Dracte; SQL server of Microsoft ; My SQL; Oracle RDBMS IBM DB2

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Lecture 3 –Detailed Database system structure/ module, DBMS adv, disadv of DBMS



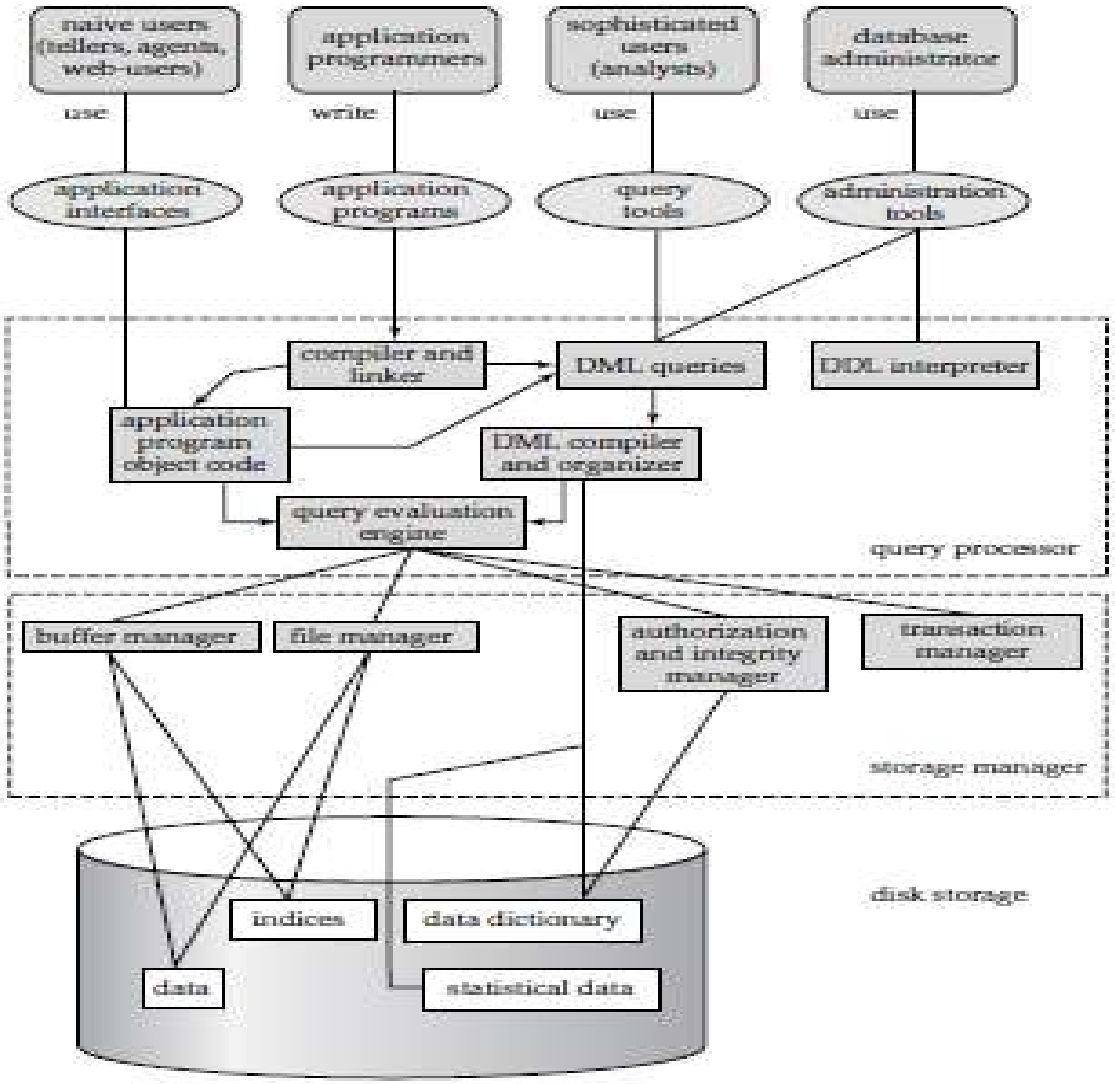


Fig. System structure/ DBMS architecture

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Fig. System structure/ DBMS architecture/ components

- 4 components for DBMS architecture:

1. Query processor
2. Storage manager
3. Transaction management
4. Different users



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- Data base user's are at top as Naive users, Application programmers; sophisticated users; specialized users/DBA { Database Administrator}.

a) Naive users:- Naive users are users who interact with the system using application programs that have been developed previously.

- The typical graphical user interface for naive users is a kind of form interface, where the user can fill in appropriate fields of the form.
- Naive users can read reports generated from the databases. They are also called as parametric users. eg.:- **Railway's ticket Booking** users; clerks in any Banks is a naive users because they don't have any DBMS knowledge but they still use the database & perform their given task.

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b) **Application Programmers** - Application programmers are responsible for writing application programs that use database.

- Application programmers are developers. or computer professionals who writes application programs.
- Programmers **develop user interface** using any preferred language like Visual Basic (VB), COBOL, FORTRAN etc.

c) **Sophisticated users** :- sophisticated user's interact with application without writing programs by using a database query Language. This query will be solved by query processor.

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- Query Processor breaks down each DML (Data Manipulation Language) query into instructions that can be understood by storage Manager.
- These user could be **engineers, scientists, Business Analyst**; who are familiar with Database. They can develop their own database applications according to their requirements:
- they don't write the program code but they interact with database by writing SQL queries directly through query processor. Eg- Analyst who submit queries to explore the data in DB.

d) Specialized Users : Creates the actual database & implements technical controls needed to enforce various policy decisions.

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- specialized users are the users who develop / works on very complicated database applications. eg:- computer Aided Design system; Data storage system with complex data type like audio; graphical data etc.; share market data [Graphs & Numbers], share Market Analysts.
- Query Processor:
- it interprets the requests (queries) received from end user via an application program into instructions, it also executes the user request which is received from the DML compiler.
- Important parts of the query processor are as follows:
- i) DDL interpreter
- ii) DML compiler
- iii) Query Evaluation engine.

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i) DDL interpreter : it processes the DDL statements **into set of table** containing meta data (data about data).The DDL (Data Definition Language) interpreter in a DBMS is used to process and execute commands that define and modify the structure of the database, such as creating tables, defining relationships, and setting constraints.

- it will interprets DDL statements & fetch the definitions in the data dictionary.

ii) DML Compiler : This processes the DML statements into **low level instructions** (machine Language), so that they can be executed. A query can usually be translated into any of the alternative evaluation plans for same query results. DML compiler will select best plan for query optimization.

- The DML (Data Manipulation Language) compiler in a DBMS is used to process and execute commands that manipulate the data stored in the database, such as inserting, updating, deleting, and retrieving records.

iii) Query Evaluation Engine : This engine will execute low-level instructions generated & by the DML compiler on DBMS.



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B. Storage Manager / storage Management : A storage manager is a program module which acts like interface between the data stored in Database & queries received. it's also called as Database control system

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- The data is stored on the disk using the file system.
- it maintains the consistency & integrity of database by applying the constraints & executes the DCL statements.
- it's responsible for **storing; retrieving. & update data** in database.
- The storage manager translates various database language statements into low level file system Commands.
- The storage Manager **components** includes :

  i) Authentication & Integrity Manager

  ii) Transaction Manager

  iii) File Manager

  iv) Buffer Manager.

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i) Authentication & Integrity Manager : it ensures role-based access control, i.e. checks whether the particular person is privileged to perform the requested operation or not.

- also, it checks the integrity constraints when database is modified.

ii) Transaction Manager : it controls concurrent access by performing the operations in scheduled way that it receives the transaction.

- Thus, it ensures that the database remains in the consistent state before & after the execution of a transactions.
- In system failure situations database is ensured in consistent (correct) state by Transaction Manager.

iii) File Manager : it manages the file space & the data structure used to represent information in database.

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iii) Buffer Manager : it's responsible for **cache memory** & the transfer of data between the secondary storage & main memory. it's important part of DB system, as it enables database to handle data sizes of that are much larger than the size of main memory.

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➢ **Data structure implemented by storage Manager :**

- Data files : it stores the data
- Data dictionary : stores the **metadata** about the structure of the database.
- Indices : Provide fast access to data items.

C) Transaction Management :

- A transaction is a series of small database operations that single large operation. (logical set of work).
- A transaction is started by issuing a "BEGIN TRANSACTION" together form a command. Once this command is executed the DBMS starts monitoring the transaction
- All operations are executed after "BEGIN TRANSACTION single large operation.

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- Transaction Management component will ensure the atomicity & durability properties.
- eg→ Transfer money from Account A to Account B.
- defined as entire transaction will happen at once or doesn't happen at all. There is no midway i.e. Transaction do not occur partially.
- it involves 2 operations : i) Abort ii) commit
- **Abort -** If transaction abort; then transaction changes made to DB not visible
- **commit -** if transaction commit; then changes made to DB are visible.
- Atomicity is also known as the "All or Nothing rule!

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Adv. Of DBMS

- DBMS removes data redundancy that means there is no duplication of data in database.
- DBMS allows to retrieve the desired data in required format
- Data can be isolated in separate tables for convenient & efficient use
- Data can be easily accessed using simple query language
- Data integrity can be maintained. that means - the constraints can be applied on data & it should be in some specific range.

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- The atomicity of data can be maintained that mean if some operation performed on one particular table of database, then change must be reflected For entire database.
- DBMS allows concurrent access to multiple users by using synchronization technique
- The security policies can be applied to DBMS to allow the user to access only desired part of database.

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Disadv of DBMS

- Complex design - Database design is complex, difficult & time consuming.
- Hardware & Software cost :- Large amount of investment needed to set up required to set up the required Hardware or to repair software failure.
- Damage Part - If one part of database corrupted or damaged, then entire database may get affected.
- Conversion cost :- If the current system is in conventional. file system & if we need to convert it to database systems then large amount of cost is incurred

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Thank You



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